

# Akshat Sharma

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**Summary** — IoT Engineering student with professional internship experience in embedded systems, computer vision, and machine learning. Skilled in hardware-firmware co-design, real-time data acquisition, and predictive modelling for safety-critical applications.

## Education

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| <b>VIPS-TC</b><br><i>BTech in Industrial Internet of Things</i>       | 2024 – 2028 |
| <b>S.R.DAV Public School</b><br><i>Secondary and Higher Secondary</i> | 2022 – 2024 |

## Skills

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| <b>IoT Hardware</b> Arduino, ESP32, MPU-6050, MQ-7<br>Sensors, OLED Displays | Scikit-learn                                                   |
| <b>Embedded</b> FreeRTOS, C++, Sensor Fusion, Power Architecture             | <b>Languages</b> Python, C, Bash, HTML, CSS                    |
| <b>AI / ML</b> Computer Vision, YOLOv5, Roboflow,                            | <b>Platforms</b> Linux Systems, Kaggle, Arduino Framework      |
|                                                                              | <b>Tools</b> Git, GitHub Desktop, Effective Prompt Engineering |

## Experience

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| <b>Innovative Intelligent Internet of Things</b><br><i>IoT Engineer Intern</i><br><u>Screw Placement Detection (Roboflow)</u> <ul style="list-style-type: none"><li>Learned the fundamentals of Roboflow and trained a <b>computer vision model</b> to accurately detect various screw placements on a manufacturing surface.</li></ul> <u>Energy Consumption Forecasting (Kaggle)</u> <ul style="list-style-type: none"><li>Used a California housing electricity dataset to build a <b>predictive regression model</b> for forecasting daily energy usage, performing feature engineering and cross-validation.</li></ul> | June 2025 - July 2025 |
| <b>Summer Internship – IIT Roorkee</b> <ul style="list-style-type: none"><li>Completed an intensive 7-day training program covering <b>Cyber Security and AI/ML</b>.</li><li>Gained hands-on exposure to ethical hacking, network security fundamentals, and supervised machine learning algorithms through guided lab sessions.</li></ul>                                                                                                                                                                                                                                                                                  | July 2025             |

## Projects

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| <b>Autonomous Field Deployment Suitability Analyzer (AFDSA)</b><br><i>ESP32, MPU-6050, C++, FreeRTOS</i> <ul style="list-style-type: none"><li>Built a low-cost autonomous robot that surveys outdoor terrain and computes a real-time <b>Deployment Suitability Score (DSS)</b> by fusing vibration, tilt, temperature, and humidity data across a grid traversal pattern.</li><li>Engineered a <b>two-rail power architecture</b> reducing motor-induced IMU noise by <math>\sim 8\times</math>; validated <b>3-class terrain classification</b> (Suitable/Marginal/Unsuitable) with 100% accuracy across 6 nodes on concrete, gravel, and soft soil.</li></ul> | March 2026   |
| <b>IoT-Based CO Level Monitor</b> <ul style="list-style-type: none"><li>Built an ESP32 + MQ-7 carbon monoxide monitor with real-time sensor calibration and OLED data visualization; co-designed hardware circuitry and firmware in a 6-member team.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                    | July 2025    |
| <b>Real-Time Document Dependency Graph and Impact Analyzer</b> <ul style="list-style-type: none"><li>Engineered a <b>Python-based Document Watcher</b> using watchdog to trigger live dependency graph updates; automated multi-document impact analysis in a team of 5.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                | October 2025 |
| <b>Asm-runner – Bash, GCC, x64 Assembly</b> <ul style="list-style-type: none"><li>CLI tool to compile and execute x64 assembly programs via GCC with GDB debug mode, argument parsing, and input validation.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                            | April 2026   |

Participation & Competitions

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| <b>Cravix – AMD Slingshot Ideathon – 3rd Place</b>                                                                                                                                       | <b>May 2026</b> |
| – Built an AI-powered recipe app (Next.js 15, TypeScript, Gemini 2.0 Flash) with dynamic AI goal generation, tradeoff sliders, and resilient API retry logic; deployed on Vercel.        |                 |
| <b>AI-Based Rockfall Prediction and Alert System – SIH</b>                                                                                                                               | <b>2025</b>     |
| – Built an ML pipeline analysing real-time vibration, moisture, and pressure sensor data to predict slope instability; integrated automated SMS/Email/onsite alerts on threshold breach. |                 |
| <b>Vibe Clash – GGSIPU East Campus Web Contest</b>                                                                                                                                       | <b>2025</b>     |
| – Built <b>NyaySetu</b> , a legal-tech platform (React, TailwindCSS, Redux) with an AI chatbot, document analysis, and lawyer directory; recognised for design and functionality.        |                 |

Position of Responsibility

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| <b>Brands and Communication Member – Entrepreneurship Cell</b>                                                     | <b>2025</b> |
| – Produced and edited viral and professional reels, significantly boosting the club’s online reach and engagement. |             |